Identifying Anomalous Rent Behavior in Atlanta

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1 Introduction

1.1 Problem Statement

In 2024, multiple states filed antitrust lawsuits against RealPage, a property management software company. They allege that the firm's access to proprietary information and algorithmic pricing creates information asymmetry that unduly harms renters throughout the USA. That is, by contracting with competing properties, rent rates are detached from ordinary market pressures. RealPage works with a high percentage of large apartment complexes throughout Atlanta.

The goal of this project is to predict the market monthly rent of apartments within Atlanta by using publicly available data. In accomplishing this goal, the model may uncover over- and undervaluation of housing, nonlinearities in pricing decisions, and fair rent rates. Ideally, this model can serve as a starting point for landlords to value their properties without violating antitrust laws. Moreover, the project seeks to identify clustering in rent prices. Such may serve as a means to identify systematic differentials in housing prices, which may be influenced, in part, by RealPage's algorithmic pricing.

The project will assess a litany of predictive models using basic, publicly available data to estimate monthly rental prices in Atlanta. Then, after selecting the best model, the test data will be clustered using a Gaussian Mixture Model. Then, the clustered data will be examined along with the respective residuals from the prior predictions, and statistical analysis via ANOVA and Tukey's HSD will reveal pairwise differences in residuals among clusters.

1.2 Data Sources

The data used in this project comes from UC Irvine's ML Repository. The dataset is composed of 100,000 entries, of which 1,506 are in Atlanta, our city of interest. There are 22 total features, though not all will necessarily be included in the model. The 22 features are as follows:

ID	category	title	body	amenities	bathrooms
bedrooms	currency	fee	has photo	pets allowed	price
price display	price type	square feet	address	city name	state
latitude	longitude	source	time		

In the instance that non-binary categorical variables are included in the model, Bayesian target encoding will be used to prevent the introduction of false ordinalities.

2 Data Processing

The process of cleaning the dataset began with extracting the subset of rows within Atlanta and ensuring that price is recorded on a monthly basis. Then, redundant features like "state" and "city" were dropped. Further, variables, such as "body," which represented a text description of the property, were dropped because they were not able to be encoded and often contained information that was captured by other features. Finally, features that contain little informational value and are unique to each unit, like "time," "id," and "address," were dropped.

Once the dataset was refined, the "amenities" column was parsed into a list of each amenity type, and the data was distributed into training and testing sets. The training set contained 80% of the data, while the testing set contained the other 20%. This, along with the remaining categorical columns, was then target encoded, with training prices being the target variable, using two-fold cross validation. Two-fold CV was chosen because five-fold was infeasible regarding certain facets of the data.

After processing, the resulting data contained 10 predictors and $1,249$ entries, split between two sets. The predictors were as follows: "amenities," "bathrooms," "bedrooms," "fee," "has_photos," "pets_allowed," "square_feet," "latitude," "longitude," and "source."

2.1 Predictive Models

2.1.1 Neural Network

We used an artificial neural network with three hidden layers of 256, 128, and 64 nodes each. Between each layer, ReLU, batch normalization, and dropout were used to introduce nonlinearity, improve convergence, and reduce overfitting respectively. After 500 training epochs, the resulting mean-squared error loss was $261,161.7$, and the overall predictive correlation on the test set was poor, as seen in Figure 1. Likely, the features in the dataset were not diverse enough for the ANN to work properly.

2.1.2 LASSO Regression

Linear regression with L_1 regularization, also known as LASSO regression, was employed to explore the linear function space. Five-fold cross-validation (CV) generated an optimal

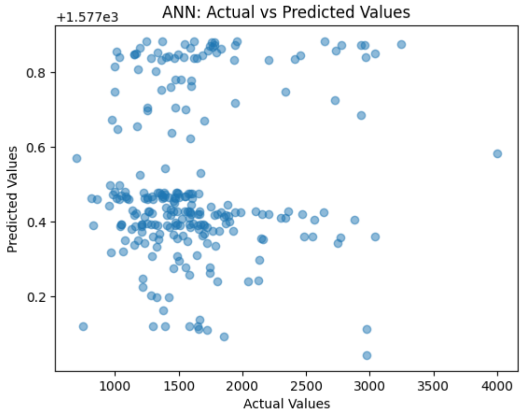

Figure 1: Artificial Neural Net actual vs predicted values on training data set.

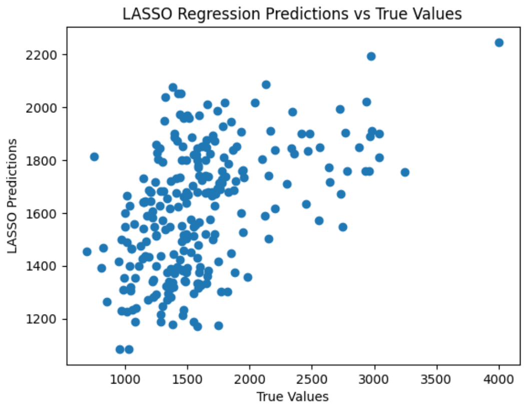

Figure 2: LASSO actual vs predicted values on training data set.

hyperparameter $\alpha \approx 1.018$, and the CV MSE was $190,954.21$. Inference MSE performance was $199,920.24$, which implies generalization is consistent with that of cross-validation. The relative success of this model is likely due to simple parameters and a small amount of training samples. Even with regularization, more expressive functions may overfit the data and generalize poorly. The predictive correlation for LASSO is demonstrated in Figure 2, and it reveals a better relationship between estimates and the true values.

2.1.3 XGBoost Regression

Finally, an XGBoost regression was implemented to strike a middle ground between the expressiveness of ANN regression and the simplicity of LASSO. Once again, five-fold cross-validation was employed, and the resulting hyperparameters achieved a CV MSE of $178,674.57$ and an inference MSE of $192,287.46$. That is, XGBoost outperformed the other models during both cross-validation and the hold-out set. The correlation between predictions and true values for the boosted algorithm, shown in Figure 3, highlights a marginally stronger performance than LASSO regression.

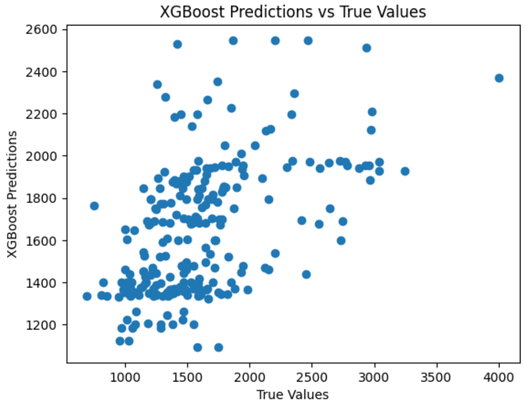

Figure 3: XGBoost actual vs predicted values on training data set.

Hyperparameter:	Value:
colsample_bytree	-1112.276
learning_rate	-196.275
max_depth	-282.412
n_estimators	1112.276
subsample	916.001

2.2 GMM Clustering

After generating the residuals from the XGBoost Regression, the test data was clustered using a Gaussian Mixture Model. The test data included all of the features and prices of the test set, and after cluster assignments were complete, the residuals column was appended. Training data was excluded because the training residuals are a function of regularization as opposed to being representative of predictability. Meanwhile, the test error will be more meaningful in capturing anomalous pricing behavior. The GMM clusters were optimized using BIC. Over the course of 20 iterations, a range of clusters were evaluated and for each iteration, the cluster that produced the "best" BIC score was recoded. Among the optimal clusters, the mode was taken to be the estimate of the true number of clusters in the population. In the case of this project, four clusters were identified.

2.3 ANOVA and Tukey's HSD

An ANOVA test confirms there is a significant distance between the GMM clusters, with an f-statistic of 69.2922 and a p-value of 1.6×10^{-32}. We then used Tukey's method to find significant pairwise comparisons.

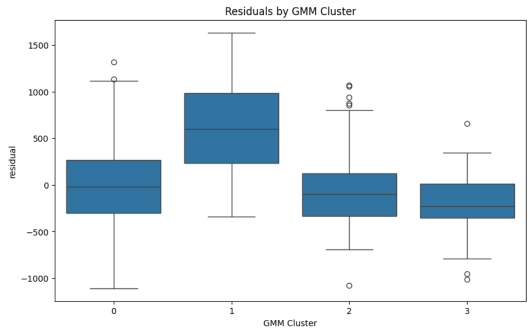

Figure 4: Gaussian Mixture Model cluster residuals.

Comparison	Statistic	p-value	Lower CI	Upper CI
(0 - 1)	-1112.276	0.000	-1403.241	-821.311
(0 - 2)	-196.275	0.186	-448.362	55.813
(0 - 3)	-282.412	0.140	-621.760	56.937
(1 - 0)	1112.276	0.000	821.311	1403.241
(1 - 2)	916.001	0.000	747.345	1084.657
(1 - 3)	829.864	0.000	546.926	1112.803
(2 - 0)	196.275	0.186	-55.813	448.362
(2 - 1)	-916.001	0.000	-1084.657	-747.345
(2 - 3)	-86.137	0.795	-328.916	156.642
(3 - 0)	282.412	0.140	-56.937	621.760
(3 - 1)	-829.864	0.000	-1112.803	-546.926
(3 - 2)	86.137	0.795	-156.642	328.916

3 Conclusion

Based on the results of Tukey's HSD test, Cluster 1 is statistically distinct from the other three, while the rest are not sufficiently distinct from each other. Intuitively, there is one small group where the model significantly underpredicts and the rest of the test set that does not significantly over- or underpredict.

In the context of the initial research problem, it makes sense that there would be a group within the data that behaves fundamentally differently than a majority of the entries. RealPage contracts with most properties, especially those that are owned by large institutions, so those that are not priced by RealPage or similar firms will exhibit price mechanics that are inconsistent with the status quo. Hence, a predictive model like XGBoost would struggle to predict the price for these properties and yield a higher residual value to a statistically significant extent. While these results are not definitive, they show promise and indicate a need for further investigation. Larger datasets and more comprehensive parameters will be needed to parse the anomalous behavior identified in this project.